

CS2 Week 1

Annotated

Discrete Time Systems



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- Interests: Robotics, Controls, AI
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Exercise Classes

- Shared exercise with Haixiao and Kissan
- Theory recap
- Notes available on Kissan's website: <https://kissanv.github.io>
& on Haixiao's Polybox: <https://polybox.ethz.ch/index.php/s/AReA9YBeJQKxBDn>
- Quizzes, Examples & Old exam questions



Kissan's Website



Haixiao's Polybox

Course Schedule

Subject	Week
Intro – Discrete Time Systems	1
Diagonalization, Modal Coordinates, Contr/Obs.	2
State Feedback Pole Placement, LQR	3
State Estimation, Luenberger Obs., Kalman Filter	4
03/17 Separation Principle, LQG, Stability margins	5
Intro to MIMO Systems, Transmission Zeros	6
Nonlinear Systems	7
break	8
Discrete-time Unconstrained Optimal Control	9
Convex Optimization	10
Model Predictive Control	11
Practical Model Predictive Control	12
Robust Model Predictive Control	13
Implementation Methods	14

Exam

- 150 minutes
- A single-sheet, 2-side, A4 "cheat sheet" is allowed
- MC questions
- Mostly follow a recipe (but understanding can be beneficial)

Today

1. Discrete Time
2. Discrete Time Systems

1. Discrete Time

What is Discrete Time?



continuous time $u(t)$:

- Defined for all times "t" in a certain interval
- For real physical systems
- Behavior as seen in CSI

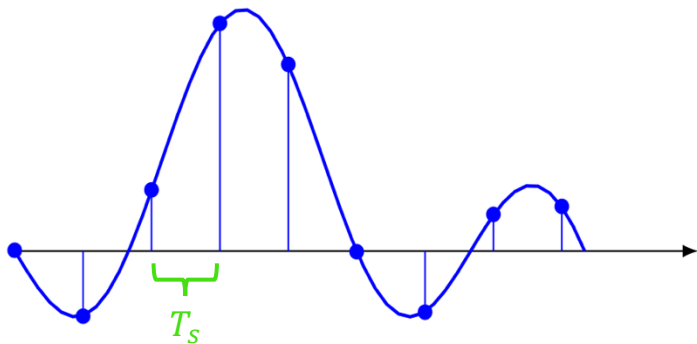
discrete time $u[k]$:

- Only defined for certain discrete steps "n"
- For computers
- One CT signal can produce many different DT signals

Sampling (CT $\rightarrow$ DT)

Remark: Sometimes we use $\omega[\text{rad/s}]$ and sometimes we use $f[\text{Hz}]$ $\omega = 2\pi f$

- A system that takes a CT and outputs a DT signal is called a sampler. It's a Linear Operation!



T_s : Sampling Time

ω_s : Sampling Frequency

$$\omega_s = \frac{2\pi}{T_s}$$

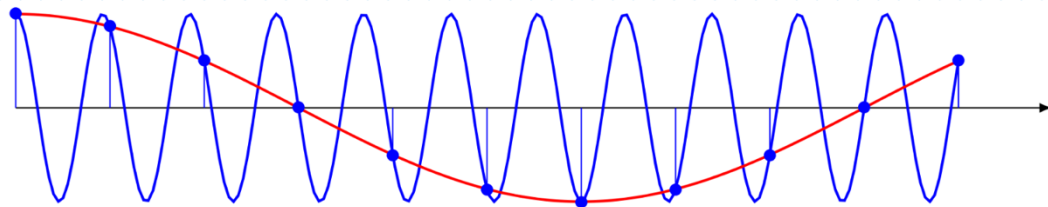
Consider $u(t) = \cos(\omega t)$

set
 $t \rightarrow kT_s$

We sample with T_s , the samples are: $y[k] = \cos(\omega k T_s), k = 0, 1, 2, \dots$

Aliasing

$$u(t) = \cos(\omega t) \longrightarrow y[k] = \cos(\omega k T_s)$$



Problem: Any frequency $\omega' = \omega + n \frac{2\pi}{T_s}$, $n \in \mathbb{N}$ will generate the same samples.

Proof:

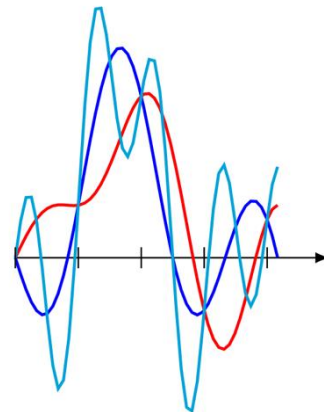
$$\begin{aligned} \text{Assume } u_1(t) &= \cos(\omega t) \\ \Rightarrow u_1[k] &= \cos(\omega k T_s) \end{aligned}$$

$$\begin{aligned} \text{Assume } u_2(t) &= \cos\left(\left(\omega + \frac{2\pi}{T_s}\right)t\right) \\ \Rightarrow u_2[k] &= \cos\left(\left(\omega + \frac{2\pi}{T_s}\right)k T_s\right) \\ &= \cos(\omega k T_s + 2\pi k) \\ &= \cos(\omega k T_s) = u_1[k] \end{aligned}$$

So $u_1[k] = u_2[k]$, even though $u_1(t) \neq u_2(t)$

Nyquist-Shannon Sampling Theorem

- Why is aliasing a problem?
 - From discrete samples alone, the original CT frequency is not uniquely determined
 - From the discrete set we could reconstruct any of the signals
 - How can we avoid aliasing?
- Nyquist-Shannon Theorem states, that we can fully reconstruct a signal if all frequencies of the sampled signal are below the Nyquist frequency:



$$f_N = \frac{1}{2T_s} [Hz] \text{ or } \omega_N = \frac{\pi}{T_s} [rad/s]$$

Holding (DT $\rightarrow$ CT)

- A system that takes a DT signal as an input and outputs a CT signal is called a hold. It's a Linear Operation!

There are multiple ways to “fill the gaps” between sampled points. In CSII, we consider the Zero-Order Hold (ZOH):

$$u(t) = u[k] \text{ for all } kT_s \leq t < (k+1)T_s$$



- The original signal was a smooth sinusoidal function, the sampling process causes a loss of information
- The Zero-Order Hold (ZOH) introduces an average time delay of $T/2$

Quiz 1

The higher the sampling time, the higher the amount of samples

A. True

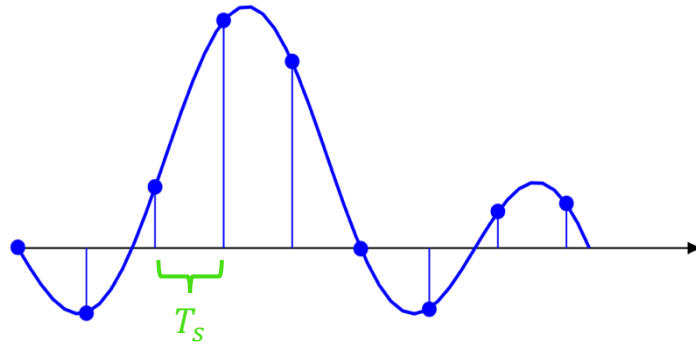
B. False

Quiz 1

The higher the sampling time, the higher the amount of samples

A. True

B. False



Quiz 2

A digital to analog converter is a type of sampler

A. True

B. False

Quiz 2

A digital to analog converter is a type of sampler

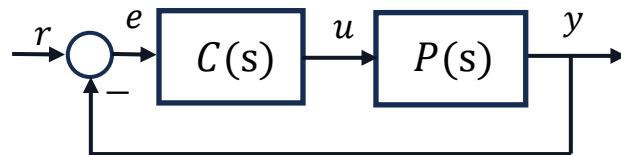
A. True

B. False

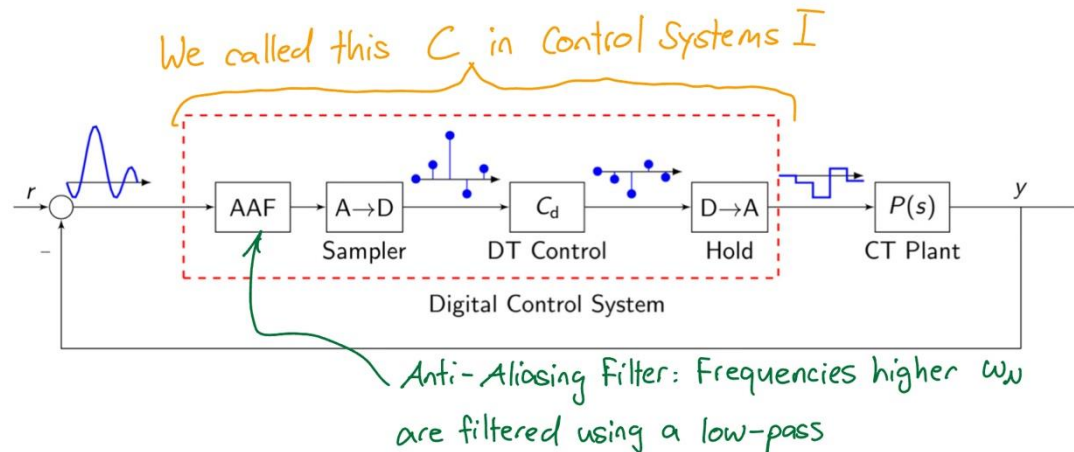
2. Discrete Time Systems

Discrete Time Systems

- But why is all of this important?
- Remember standard feedback architecture from CS1 :
- Plant P is a physical system $\rightarrow$ operates in CT
- Controller C is a computer $\rightarrow$ operates in DT



How C actually looks like:



State-Space Model

- So how can we use the control techniques that we develop in DT?

In continuous time (CT) we had:

$$\begin{aligned}\dot{x}(t) &= A_c x(t) + B_c u(t) \\ y(t) &= C_c x(t) + D_c u(t)\end{aligned}$$

In discrete time (DT) we now have:

$$\begin{aligned}x[k+1] &= A_d x[k] + B_d u[k] \\ y[k] &= C_d x[k] + D_d u[k]\end{aligned}$$

- From this, the response of the system can be evaluated as:

$$y[k] = C_d A_d^k x_0 + C_d \sum_{i=0}^{k-1} A_d^{k-i-1} B_d u[i] + D_d u[k]$$

⇒ This looks similar to the continuous response from CS1:

$$y(t) = C_c e^{A_c t} x_0 + C_c \int_0^t e^{A_c(t-\tau)} B_c u(\tau) d\tau + D_c u(t)$$

Notice: **exponentials** become **powers**

Stability

$$y[k] = C_d A_d^k x_0 + C_d \sum_{i=0}^{k-1} A_d^{k-i-1} B_d u[i] + D_d u[k]$$

⇒ The condition for stability changes:

- In DT we need $\lim_{k \rightarrow \infty} A^k = 0$

This requires all λ_i to be within the unit disk on the complex plane:

$$|\lambda_i| < 1, \quad \forall i = 1, \dots, n$$

z-Transform

- In CS1 we used the Laplace-Transform:

$$\mathcal{L}\{x(t)\} = \int_0^{\infty} x(t) e^{-st} dt$$

- Now we use a similar operation called z-Transform:

$$Z\{x[k]\} = \sum_{k=-\infty}^{\infty} x[k] z^{-k}$$

Note: In this course you don't need to perform these transforms

- Now, transfer functions can be found identically to how it was done in CT:

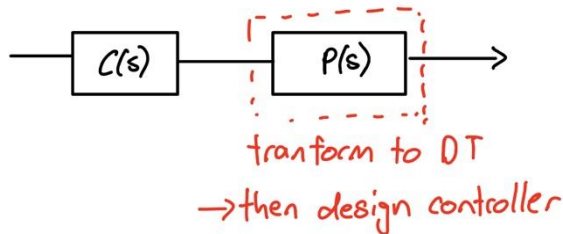
$$G(z) = C_d(zI - A_d)^{-1}B_d + D_d$$

Implementation of DT-Systems

- Assume that you have the CT system matrices of the plant. How do you design a DT Controller?
- There are two options:

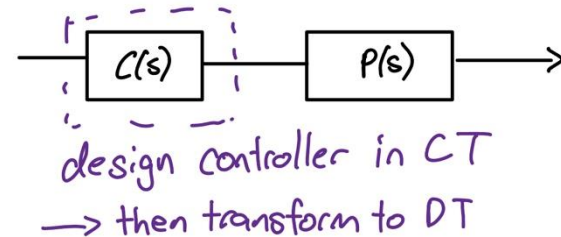
1. Direct DT design:

Transform Plant matrices to DT and design everything in DT

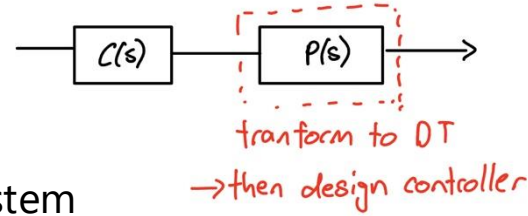


2. Emulation:

Design everything in CT and then emulate the system matrices of the controller to DT



1. Exact Discretization



- Find a DT system that behaves equivalently to the given CT system

First let's look at the transient response

$$\begin{aligned} \dot{x}(t) &= Ax(t), & x(0) &= x_0 \\ y(t) &= Cx(t) \end{aligned}$$

→
integrating

$$\begin{aligned} x(t) &= e^{At} x_0 \\ y(t) &= Cx(t) = Ce^{At} x_0 \end{aligned}$$

↓
discretizing
with $t = kT$

$$x(kT) = e^{AkT} x_0 = \underbrace{(e^{AT})^k}_{A_d} x_0$$

Applying the same reasoning to the steady-state response gives the remaining DT matrices:

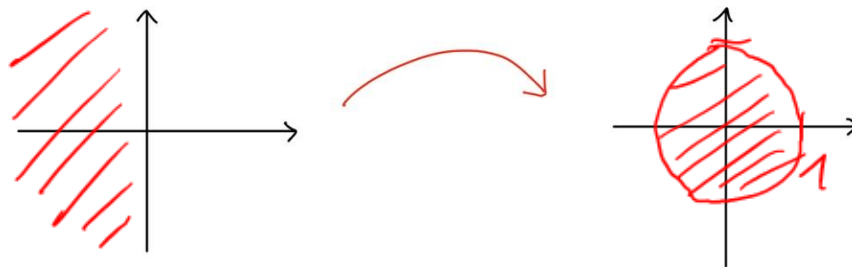
$$\begin{aligned} A_d &= e^{AT_s} & B_d &= \left(\int_0^{T_s} e^{A\theta} d\theta \right) B \\ C_d &= C & D_d &= D \end{aligned}$$

Remark: If A is invertible, then:

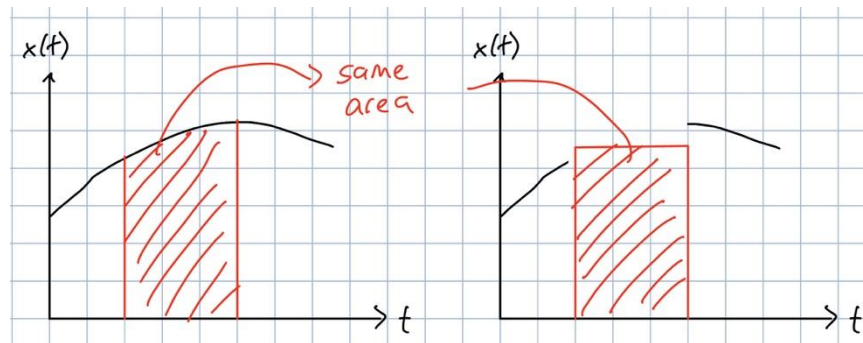
$$B_d = A^{-1}(A_d - I)B$$

1. Exact Discretization

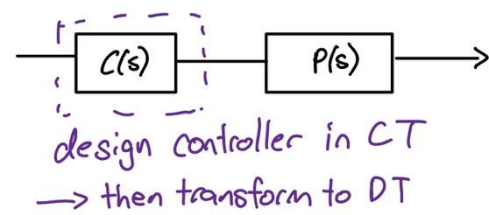
- Exact Discretization maps stable poles to the unit circle:



- Intuitively, exact discretization can be understood as an Integral:



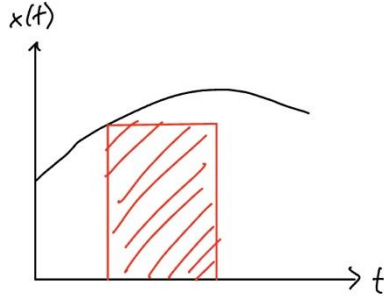
- In the frequency domain this corresponds to the substitution: $z = e^{sT_s}$



2. Emulation

- The second method emulates this procedure by approximating the integral. We will look at three methods.

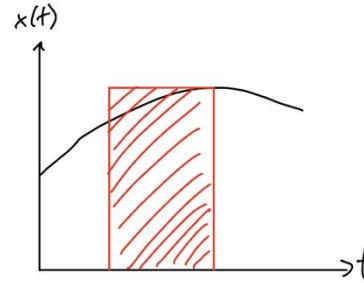
Euler-Forward



Substitution:

$$z = sT_s + 1$$

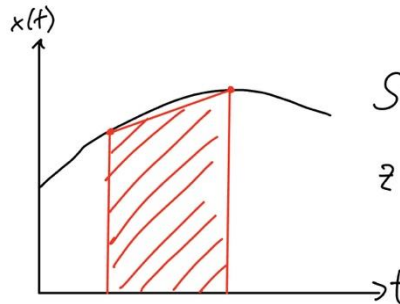
Euler-Backward



Substitution:

$$z = \frac{1}{1 - sT_s}$$

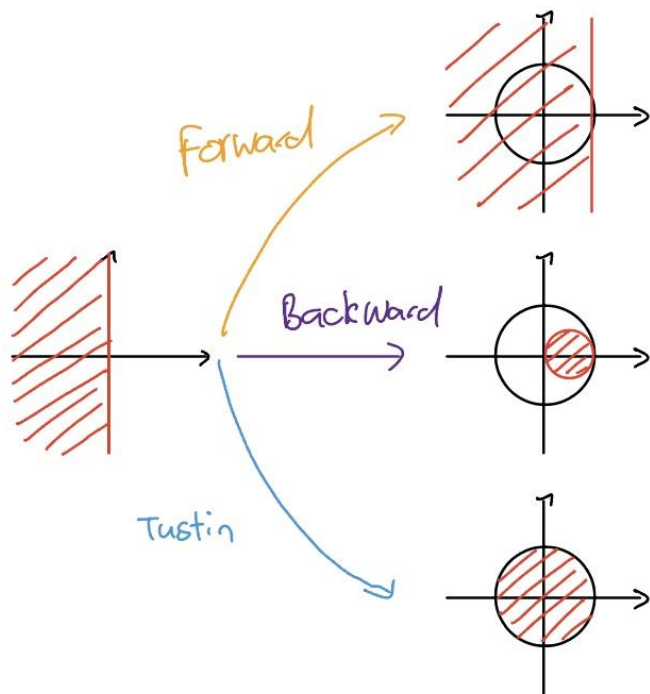
Tustin's method



Substitution:

$$z = \frac{1 + sT_s/2}{1 - sT_s/2}$$

Summary



Exact	$s = \frac{1}{T_s} \cdot \ln(z)$	$z = e^{s \cdot T_s}$
Euler forward	$s = \frac{z - 1}{T_s}$	$z = s \cdot T_s + 1$
Euler backward	$s = \frac{z - 1}{z \cdot T_s}$	$z = \frac{1}{1 - s \cdot T_s}$
Tustin	$s = \frac{2}{T_s} \cdot \frac{z - 1}{z + 1}$	$z = \frac{1 + s \cdot \frac{T_s}{2}}{1 - s \cdot \frac{T_s}{2}}$

Note: In contrast to Euler Forward, Backward Difference and Tustin's method preserve stability!

Quiz 1

Using exact discretization, where does a pole at $s = 0$ map?

A. $z = 0$

B. $z = 1$

C. $z = 2$

D. $z = 3$

Quiz 1

Using exact discretization, where does a pole at $s = 0$ map?

A. $z = 0$

B. $z = 1$

C. $z = 2$

D. $z = 3$

- In the frequency domain this corresponds to the substitution: $z = e^{sT_s}$

Quiz 2

Using forward difference, a stable pole in CT is also stable in DT

A. True

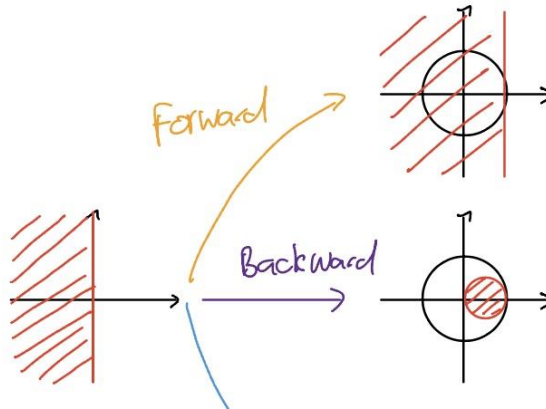
B. False

Quiz 2

Using forward difference, a stable pole in CT is also stable in DT

A. True

B. False



Old Exam Question

Problem 1 (Discrete-time Control Loop)

You are given the control loop depicted in Figure 1.

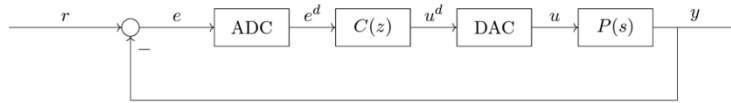


Figure 1: Discrete-time control loop.

- a) Assign the 6 signals of the control loop (r, e, e^d, u^d, u, y) to the signals drawn in Figure 2.

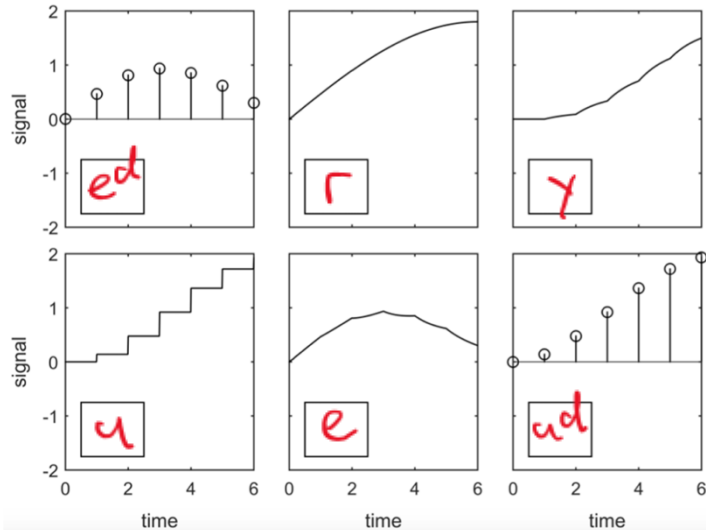


Figure 2: MC for the given signals

Tips for Problem Set 1

- 1: Check if the all frequencies of the signal are below the Nyquist frequency $f_N = \frac{1}{2T_s}$ or $\omega_N = \frac{\pi}{T_s}$
- 2: What condition do the eigenvalues of a DT system have to fulfill?
- 3: Use the equation for the system response & transfer function.
The DC gain is given by $G(z = 1)$
- 4: Apply the substitution for Tustin's method
- 5: Use the formulas for converting CT system to DT also note that $A_d = I + A_c T_s + \frac{(A_c T_s)^2}{2} + \dots$

Feedback?

Too fast? Too slow? Less theory, more exercises?
I would appreciate your feedback. Please let me know.